A Socioeconomic Informative Dashboard for Migrations and Relocations

In partial fulfillment of the requirements for the course
Visualizations and Data Analysis
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Submitted by:

Pangilinan Jr. Bienvenido B.
Pasion, Al Twain D.
Torres, Carl Andrei D.

Submitted to:

Engr. Jimlord Quejado

Technological Institute of the Philippines - Quezon City

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I. Problem Background and Validation

- The problem is present around the world, understanding it is critical to debunk the misconception that a low-cost of living translates to an easy life, financially. By using macroeconomic datasets, this study transforms our perspective on housing from a subjective point of view to a more objective reality. This demonstrates how significant housing costs are in dictating the overall quality of life for a nation's citizens.

a. The Problem

- The Global Housing Affordability Crisis: Canvassing only through cheap daily housing costs creates a false narrative of affordability. By evaluating the macroeconomic data presented, this project aims to demonstrate how the rise of Property Price-to-Income ratios severely depreciates the national Quality-of-Life scores, exposing the truth that low consumer costs make for severe and systematic housing traps.

WHO MIGHT BE AFFECTED BY THIS PROBLEM:

- **Prospective Migrants and First-Time Buyers**: People who are looking to relocate or buy a house who need to evaluate the financial viability.
- **Low to Middle-Income Families**: Working households who are vulnerable to income allocation traps, where housing prices negatively impact spending power for health, safety, and quality of life.
- **Urban Workforce & Remote professionals**: Workers who research commute times, regional purchasing power against local living conditions, and spatial misalignment.
- **Socioeconomic Researchers & Analysts**: Analysts monitoring and evaluating macro-level international quality-of-life trends and structural housing expenditures across multiple countries.

b. Problem Validation

- According to Yanotti et al. (2023), across many nations, the gap between house prices and household incomes has widened internationally. This demonstrates that housing affordability is a systemic, structural economic issue rather than a personal financial failing.

II. The Project

- This project utilizes statistical data visualization to map out the correlation between a nation's housing affordability and its citizens' overall quality of life. By using the effective graphs, the goal of this project is to visually rebrand the belief that moving to a country with a "low cost of living" can help with financial struggles. Thus proving that local purchasing power and housing raptors are one of the main factors of livability.

a. Dataset

ORIGIN OF THE DATASET:

The Dataset comes from Kaggle.com .

CONTRIBUTORS:

Ahmed Mohamed

TYPE OF DATA:

This dataset is composed of ***numerical scores*** and ***descriptive categories***.

Numerical Scores:

- Purchasing Power Value
- Safety Value
- Health Care Value
- Climate Value
- Cost of Living Value

Descriptive Categories:

- Purchasing Power Category
- Safety Category
- Health Care Category
- Climate Category

TOTAL NUMBER OF ENTRIES:

Total Rows: 236 Countries (each row represents each country)

Total Columns: 19 Attributes (1 entity identifier column, 9 numerical score columns, and 9 qualitative category columns)

RELEVANT ATTRIBUTES:

CORE ENTITY:

- **Country** -> The specific country or nation being evaluated

KEY ATTRIBUTES TO HOUSING AFFORDABILITY ANALYSIS:

- **Property Price to Income Value *(Numerical)*** -> Measures how affordable housing is relative to the average salary. Lower scores indicate better housing affordability; higher values means that the housing is severely expensive relative to the average earnings.

- **Property Price to Income Category *(Categorical)*** -> Measures how affordable housing is relative to the average salary which are qualitatively measured ('Low', 'Moderate', 'High', 'Very High').
- **Quality of Life Value *(Numerical)*** -> Total index that measures the overall living conditions.
- **Quality of Life Category *(Categorical)*** -> Qualitative rating of the overall living standards.
- **Cost of Living Value and Category** -> Overall cost of the goods and services that are relative to the indices.
- **Purchasing Power Value and Category** -> The Purchasing Capability of the citizens.

SUPPORTING QUALITY-OF-LIFE ATTRIBUTES:

- **Safety Value & Category** -> National safety and crime rate metrics.
- **Health Care Value & Category** -> Assessment of the healthcare quality, responsiveness, and infrastructure.
- **Traffic Commute Time Value & Category** -> Average time spent when commuting which also reflects the urban sprawls and traffic congestion.
- **Pollution Value & Category** -> Environmental cleanliness and pollution exposure index.
- **Climate Value & Category** -> Climate favorability score.

CLEANING METHOD AND MODIFICATIONS

The group used cleaning via Python in Google Colab to isolate, clear, and modify values such as:

- Error values,
- Missing values,
- Unwanted characters, and
- Proper spacing

b. Data Dashboard

- Purpose: The dashboard serves as a visual aid decision-support platform for prospective migrants, families, workers, and socioeconomic analysts to evaluate international livability and financial viability.

- Core Metrics: This visualizes the key macroeconomic indicators including Property to Price Income Rations, Purchasing Power Indexes, Cost of Living Categories, and overall Quality of Life Scores.

c. Project Description

Overall Description of our Project:

Our project is creating a Socioeconomic Informative Dashboard for Migrations and Relocations which can serve the migrants an interactive visualization and decision support tool that is designed to map out the storyboard that correlates every Nation's housing affordability and its quality of life. By translating the macro-economic datasets that we've gathered into a clear graphical dashboard, it can make the users make informative migration choices and challenge the common misconception that a low-cost nation automatically offers a high quality of financial and social life.

Key Objectives:

- This project dashboard uses a chart to show how the housing affordability affects a country quality life and it also help the people to see which countries that are safe and livable, not just cheap

- The dashboard shows how the cost of housing shapes the quality of life in different countries

- To show not just because it's cheap, does not mean it is an easy place to live in.

- This dashboard is a guide to help migrants or families that want for better relocation choices using simple, clear visuals.

- To analyze the dataset of 236 countries by comparing property prices related to income, purchasing power, safety, health care, and weather conditions.

III. Output

Property Price to Income Ratio VS Quality of Life Value

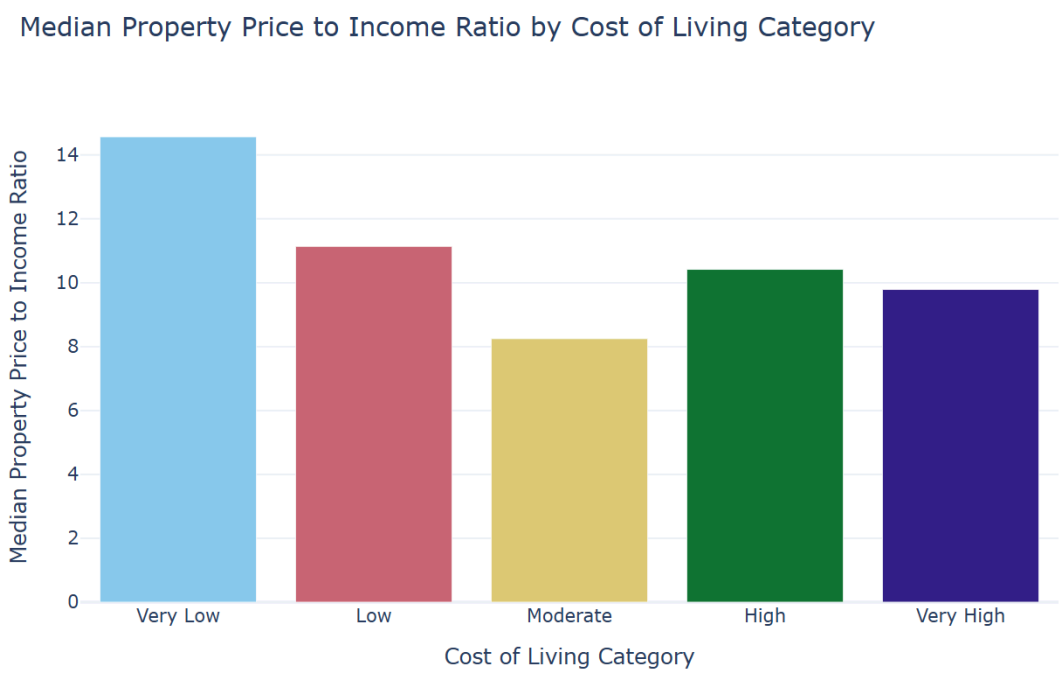


DESCRIPTION / ANALYSIS / FINDINGS

In this scatter plot with the average trend line, we can see the strong negative correlation of all the countries between a nation's Property Price-to-Income Ratio and its overall Quality of Life Index.

From our observations, as housing costs increase, it depresses the nation's quality of life. This proves that severe housing prices can directly limit the people's well-being and living conditions regardless of other regional amenities.

Average Property to Price Income Ratio by Cost of Living Category

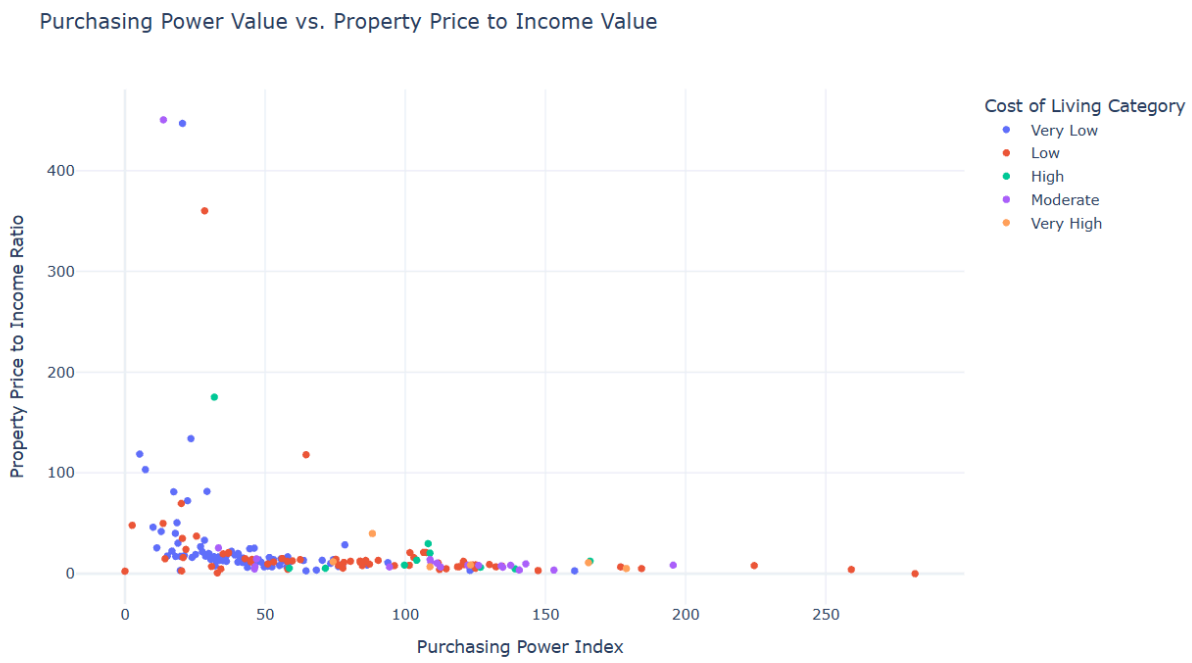


DESCRIPTION / ANALYSIS / FINDINGS

This bar chart presents the qualitative "Cost of Living" value categorized by "Very Low", "Low", "Moderate", "High", and "Very High". We can see that the very low cost of living nations suffer from the highest median property price to income ratio which exceeds the value of 14, meaning that the Cost of Living may be good but when you take into account factors such as the ratio of property price to income, it is severe.

This exposes the truth that countries with cheap everyday consumer goods often have depreciated wages relative to that of property prices. Moving to a "cheap" country doesn't mean it's a good financial decision if housing prices are very far from earnings.

Purchasing Power Value vs. Property Price to Income Value

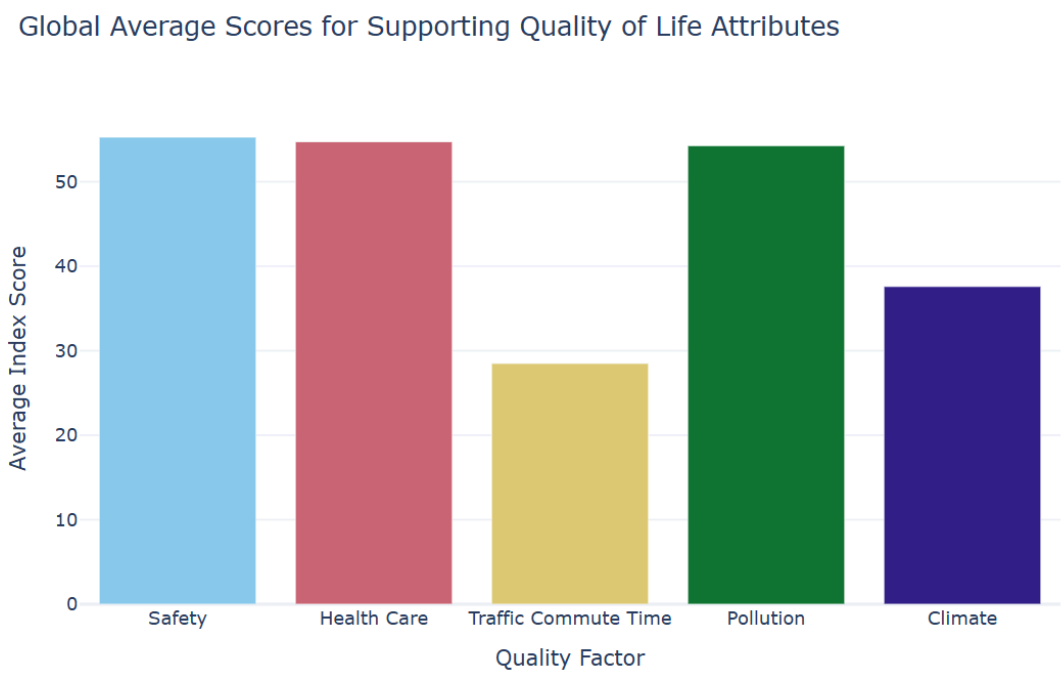


DESCRIPTION / ANALYSIS / FINDINGS

This scatter plot shows that high property price to income ratios are mostly grouped with low purchasing power. Once a nation's purchasing power Index crosses about a hundred, the property price to income ratio settles below severe levels.

A nation's purchasing power can be the "economic buffer" in determining the true livability and financial stability. It depends far more on purchasing capacity than on low surface consumer prices.

Global Average Scores for Supporting Quality of Life Attributes

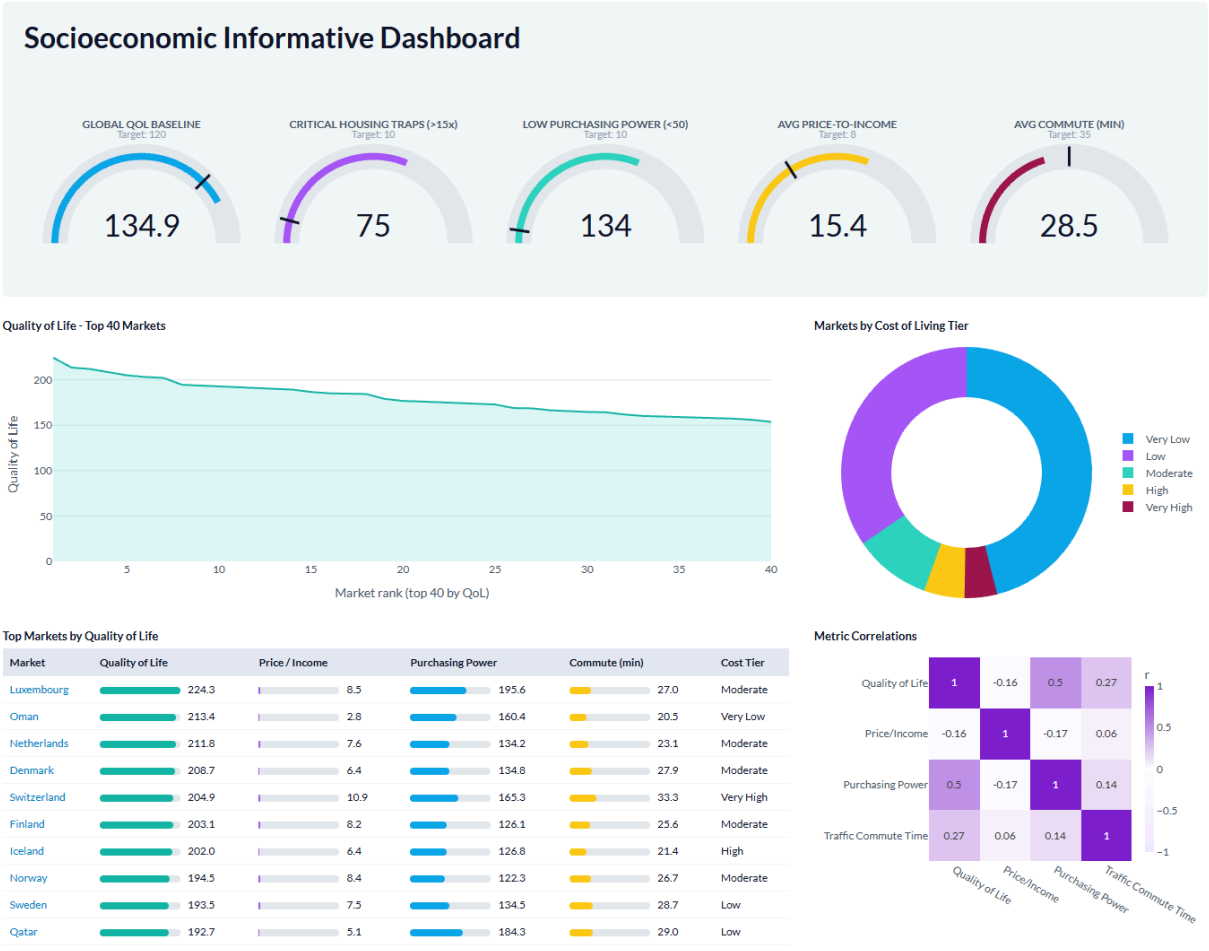


DESCRIPTION / ANALYSIS / FINDINGS

This bar chart shows that while the global average of supports is high, traffic commute time drops to 30.

It highlights the spatial mismatches that workers are forced into long commutes due to poor urban housing prices. This is one of the primary points that depresses daily quality of life, even in nations that have quality health care and safety.

SOCIOECONOMIC INTERACTIVE DASHBOARD



DESCRIPTION / ANALYSIS / FINDINGS

Our dashboard shows the average global quality of life is high at 134.9, and also many countries struggle with housing and income, while 75% of those countries have housing that costs more than 15 times of the average income and 134 countries have low purchasing power. It shows that a cheap country is not always an easy place to live when incomes cannot keep up with housing prices.

SUMMARIZATION

Our analysis proves that the traditional ‘cost of living’ metrics compared to the modern values are deeply misleading for financial and migration planning. Countries with the lowest consumer prices often present the highest housing prices relative to their income. The real factors comes from the purchasing power and lower property price to income ratios rather than lower prices alone

MIGRANTS/WORKERS/FAMILIES before our Dashboard

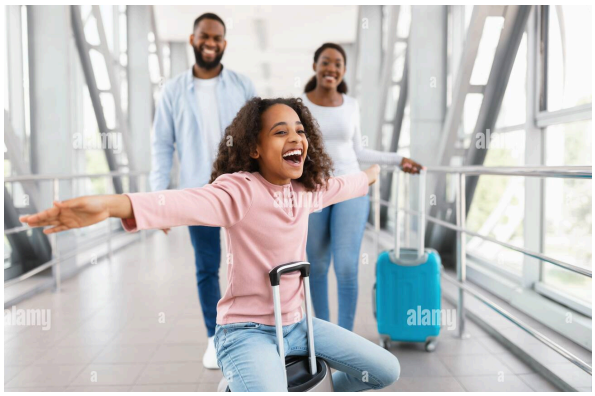




Before: "Overwhelmed by hidden property ratios & predatory housing markets (>15x)."

MIGRANTS/WORKERS/FAMILIES after our Dashboard





After: "Empowered by pre-move risk analysis and targeted caseworker support."

IV. Conclusion

As researchers, we've managed to create an Interactive Dashboard that would be useful for the planning of the migrant workers and their families on their migration journey to another country. By evaluating the global macroeconomic dataset across 236 countries into actionable graphical or visual insights, which our project successfully challenges the common misconception that a low cost nation does not guarantee a good social and financial well-being. The findings from the graphs that are used on the dashboard shows that nations categorized under a "Very Low" cost of living frequently suffer from severe Property Price-to-Income ratios, which directly depressing the overall Quality of Life scores and worsening the urban commute burdens. From our storyboard, we can see that before the Migrant Workers and their Families didn't use our dashboard, they are overwhelmed by the hidden Property Price-to-Income ratios and the predatory housing markets because they didn't researched well about the country they want to migrate in which can worsen their social and economic well being. After they've used the Socioeconomic Interactive Dashboard, they now have the knowledge of a certain country which they acquired pre-move risk analysis and a targeted caseworker support for the Migrant Workers and Families. As we conclude, this project proves that the true livability depends far more on local purchasing power and structural housing affordability than on cheap everyday consumer goods. Also, our dashboard enables individuals and organizations to conduct pre-move risk assessments, mitigate predatory housing risks, and execute long term relocation strategies that are grounded on objective economic reality of each country.

V. Development Team

- Team 6 is composed of three students from the Technological Institute of the Philippines - Quezon City who developed this project for the course Visualizations and Data Analysis. The team studied the relationship between housing affordability and quality of life across countries using the Numbeo dataset from Kaggle. All three members took part in data collection, while the remaining tasks were divided according to each member's role, Pasion led the system design, and Pangilinan and Torres handled the slides, documentation, and presentation.

a. Team Description

Pangilinan	- Data collection, PPT, Documentation, Presenter
Pasion	- Data collection, Documentation, Dashboard Presentation
Torres	- Data collection, PPT, Documentation, Presenter

- The team works as a small group in which everyone contributes to the data and each member owns a specific part of the output. Pasion focuses on how the system and dashboard are designed, while Pangilinan and Torres focus on communicating the results through the slides, the written documentation, and the final presentation. The team's goal is to present clear, data-driven visualizations that show how housing affordability shapes a country's quality of life, and to challenge the belief that a low cost of living automatically means an easier life.

b. Profile



Pangilinan Jr. Bienvenido B.

- I am Bienvenido B. Pangilinan Jr., a student at the Technological Institute of the Philippines - Quezon City (TIP-QC) and a member of Team 6 in Visualizations and Data Analysis. In this project I took part in data collection and was responsible for the PPT, the documentation, and presenting the team's work. Skills and expertise: data collection, document writing, slide preparation, and presenting.



Pasion, Al Twain D.

- I am Al Twain D. Pasion, a college student at the Technological Institute of the Philippines - Quezon City (TIP-QC). I'm currently taking the Bachelor of Science in Computer Engineering program. I'm adequate in basic C++ and Python programming languages. I value the process of learning and believe that time is only secondary when it comes to what you're passionate about. In this project, I am the lead coder, data collector, presentation maker, and debugger.



Torres, Carl Andrei D.

- I am Carl Andrei D. Torres, a 2nd Year Student at Technological Institute of the Philippines - Quezon City (TIP-QC) where I serve as the Documenter, Data Collector, Presenter, and PPT Creator of our Storyboard in Visualizations and Data Analysis. I am interested in making documents and presentations (PPT) which I'm good at. Also, I am one of the Data Collectors for our Storyboard which can be helpful for understanding the solution to our problem carefully. Also, I am one of the presenters that will present our created PPT and Storyboard which can be crucial for explaining the solution that we've created for the problem. Moreover, I've managed to create documents and presentations in recent projects from different courses. Lastly, my goal for this project is to create documents and presentations that are understandable for the executive and manage to present for us to connect our solution to the problem successfully based on our storyboard output.

VI. References

Mohamed, A. (2025). Quality of life for each country. *Kaggle.Com*.

https://www.kaggle.com/datasets/ahmedmohamed2003/quality-of-life-for-each-country

Saiz, A. (2023). The Global Housing Affordability Crisis: Policy options and Strategies. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.4402329